# Social-Aware Edge Intelligence:

# A Constrained Graphical Bandit Approach

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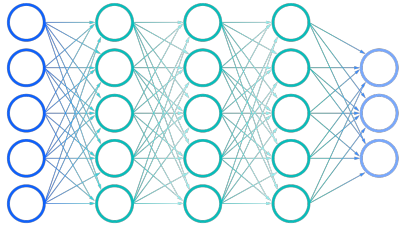
# Edge Intelligence System

**Edge** Systems

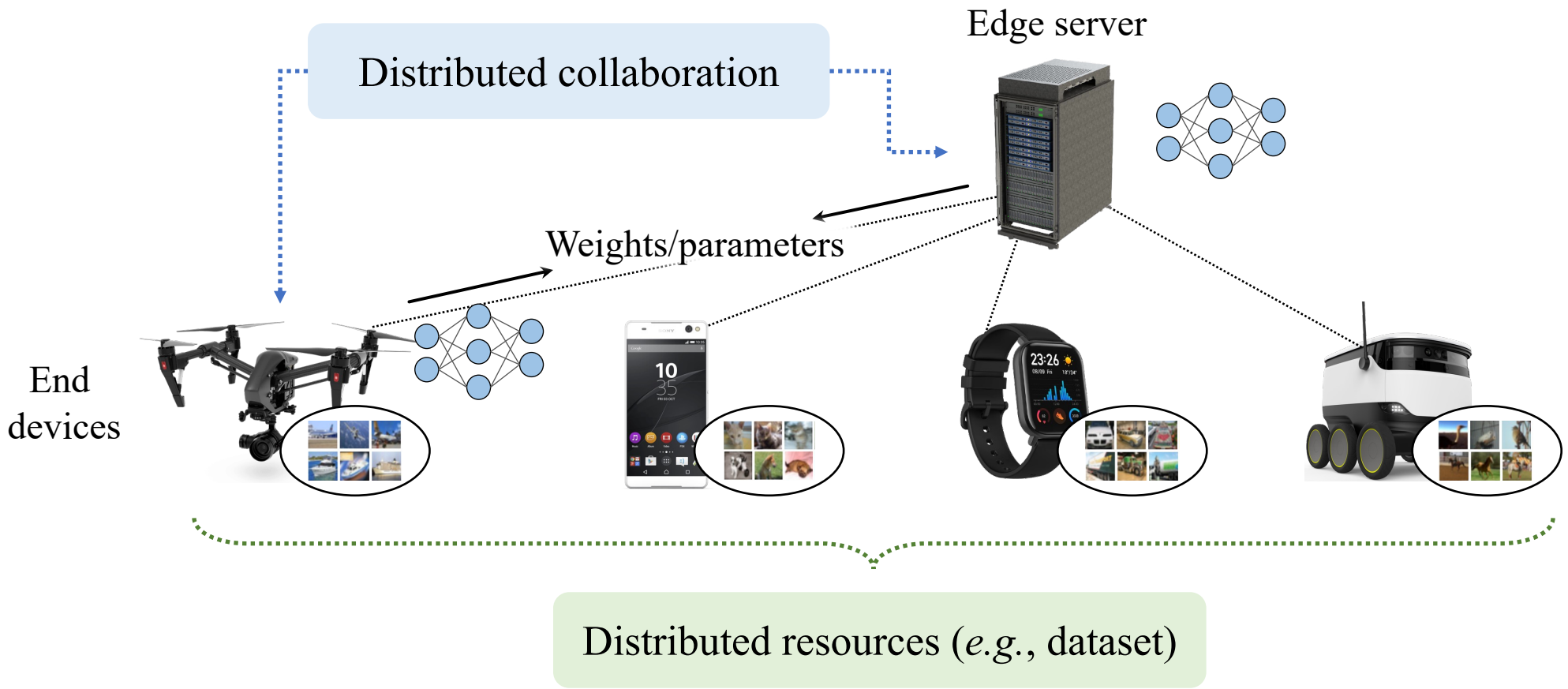
Artificial **Intelligence**

Statistical model



Neural model

# Distributed Training in Edge Intelligence System



# Device Selection Problem

- Limited connection capacity
- Multi-round selection for training

**How to select end devices in each round ?**

# A Bandit Formulation

Edge server (agent)

End devices (arms)

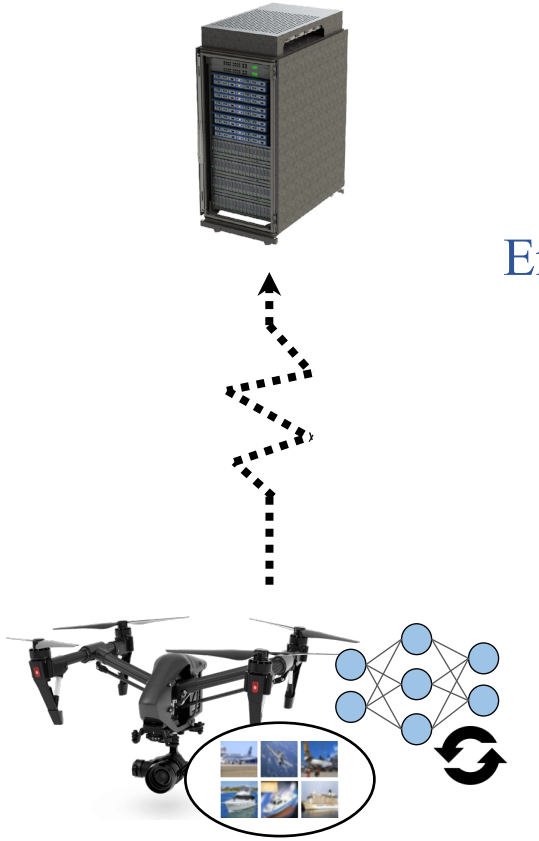
Wireless transmission

Local model training

Negative reward

Unknown and time-varying latency

# Budget Constraint



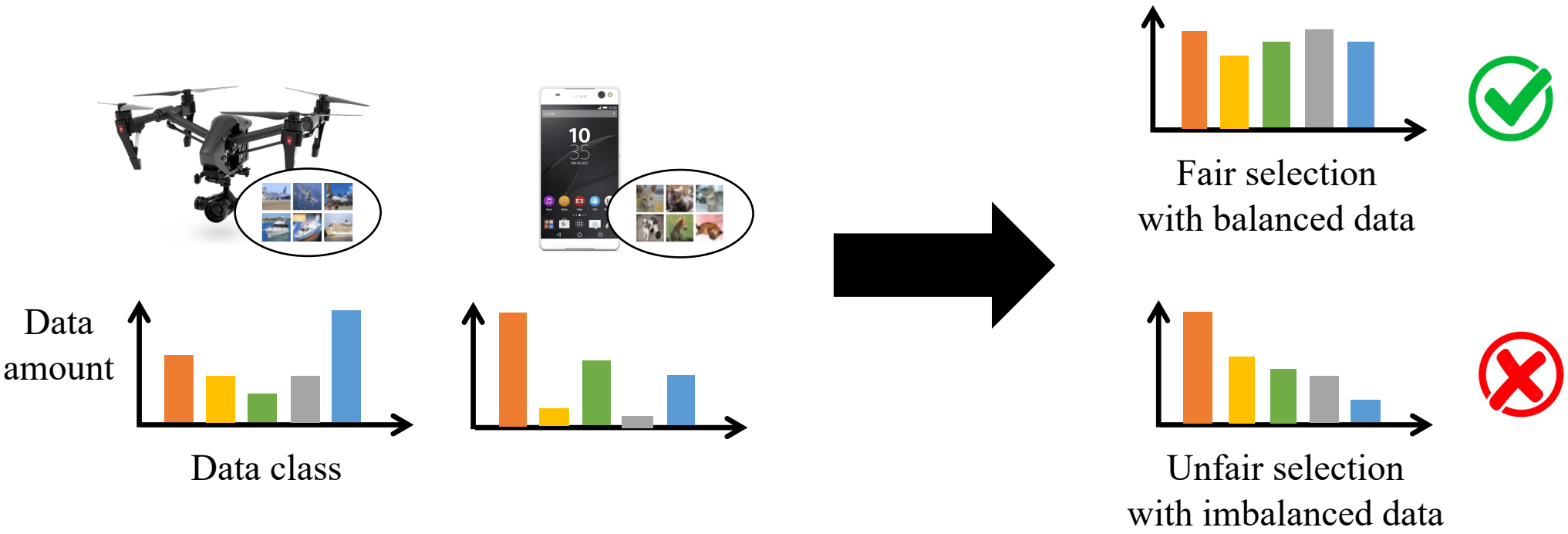
Energy consumption

Uplink transmission

Local model training

Local battery of limited capacity

# Fairness Constraint

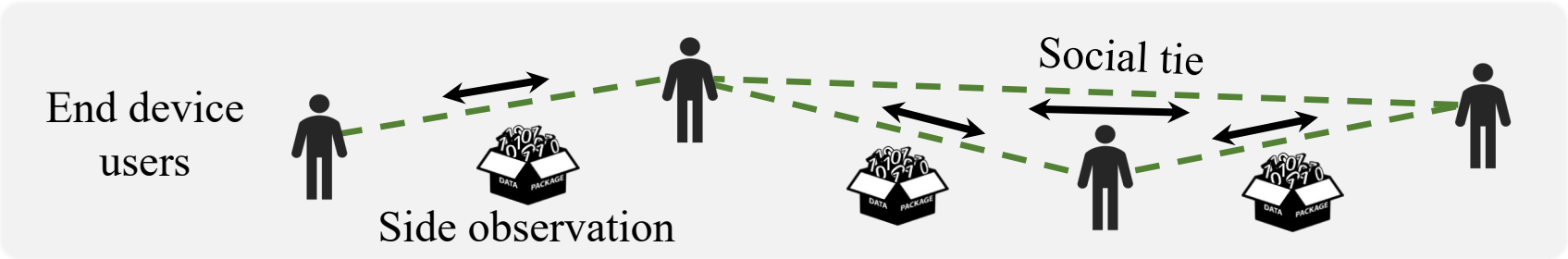


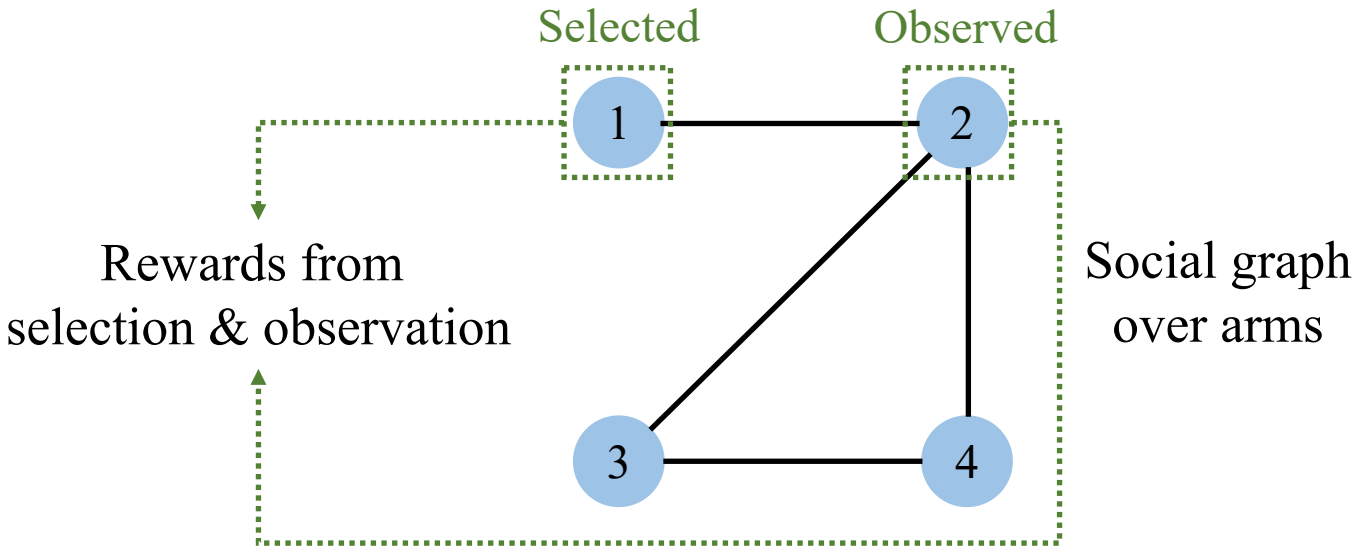
# A Constrained Optimization Problem

- ❑ maximize Reward — Latency minimization
- ❑ subject to Energy consumption ≤ a budget — Budget constraint

  Selection proportion ≥ a threshold — Fairness constraint

# Side Observation via Social Ties

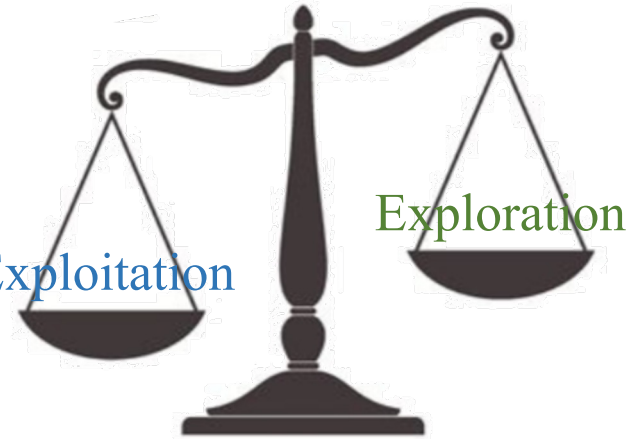


# Upper-Confidence-Bound-Style Estimation

Parameter related to the social graph $\mathcal{G}$

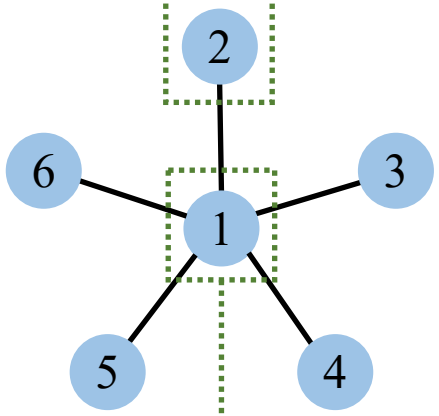
Empirical mean $+ \; c_i \sqrt{\dfrac{\log t}{\text{\# of observed \& selected times}}}$



# Social-Aware Design

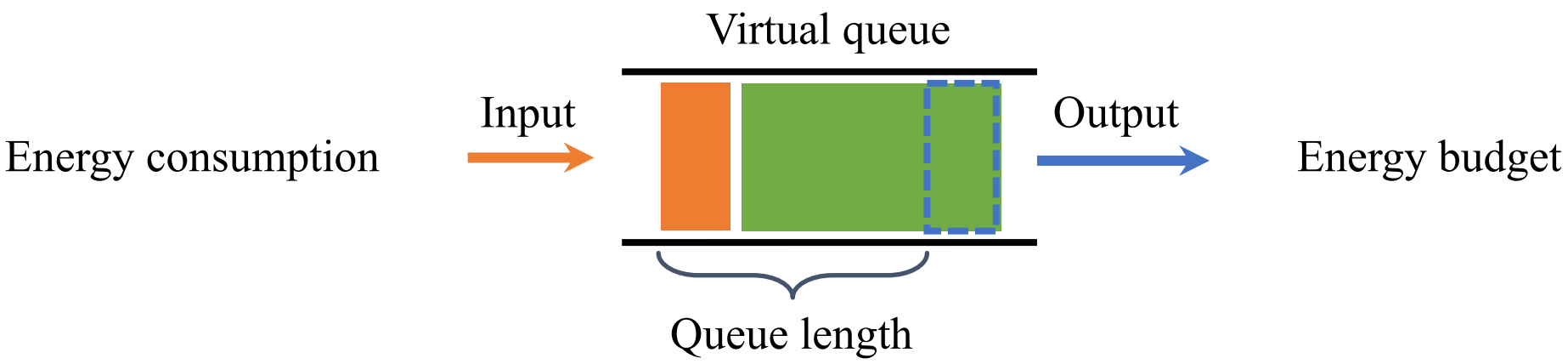
Compensate less observed arms

$$c_i \propto \frac{1}{\deg(i) + 1}$$



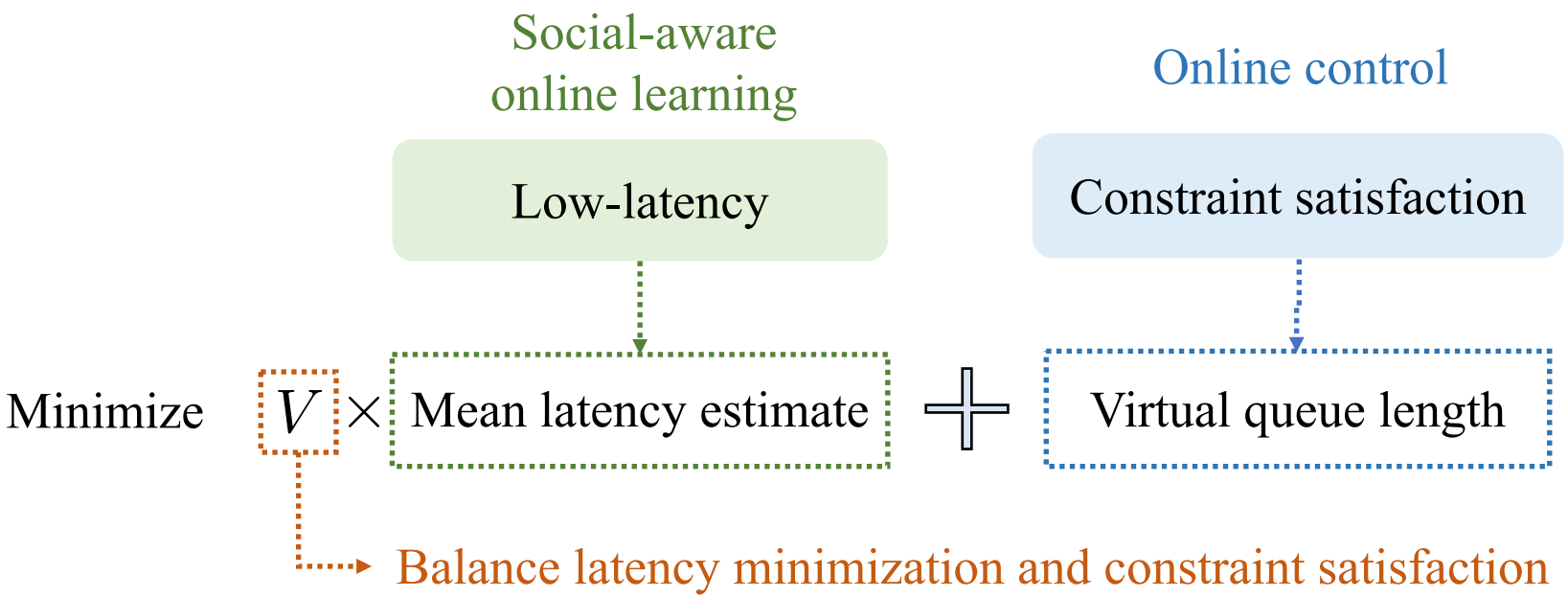

"Punish" frequently observed arms

# Online Control with Virtual Queue




Constraint satisfaction ⟷ Controlled virtual queue length

# Integrated Algorithm Design



# Regret Analysis

Online control

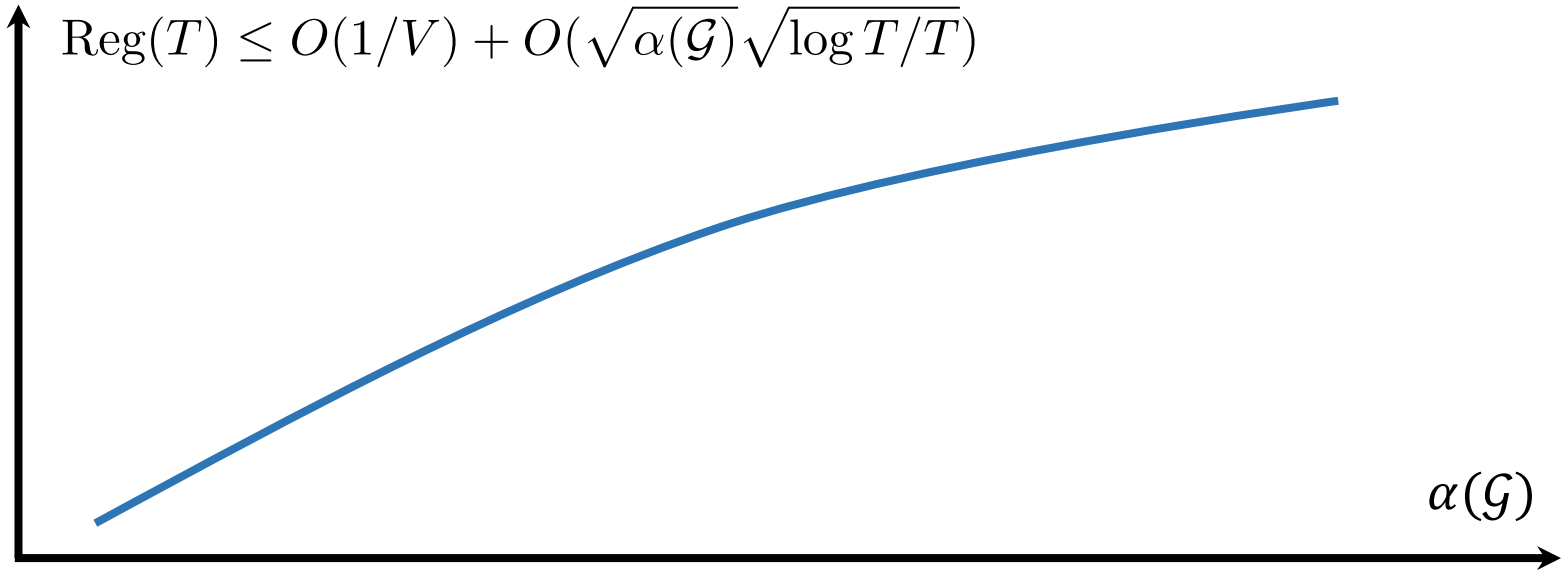
Social-aware online learning

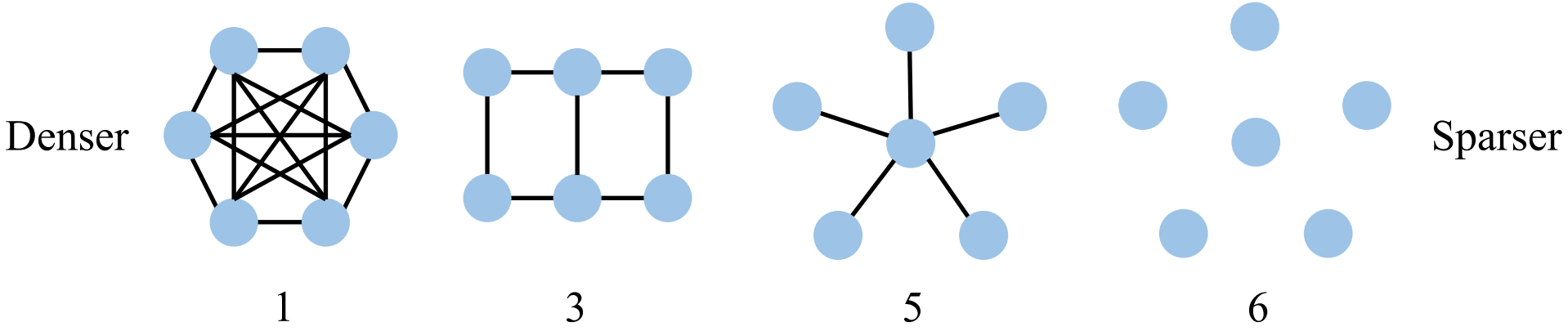
$$\mathrm{Reg}(T) \leq O(1/V) + O(\sqrt{\alpha(\mathcal{G})}\sqrt{\log T/T})$$

Independence number of the social graph $\mathcal{G}$

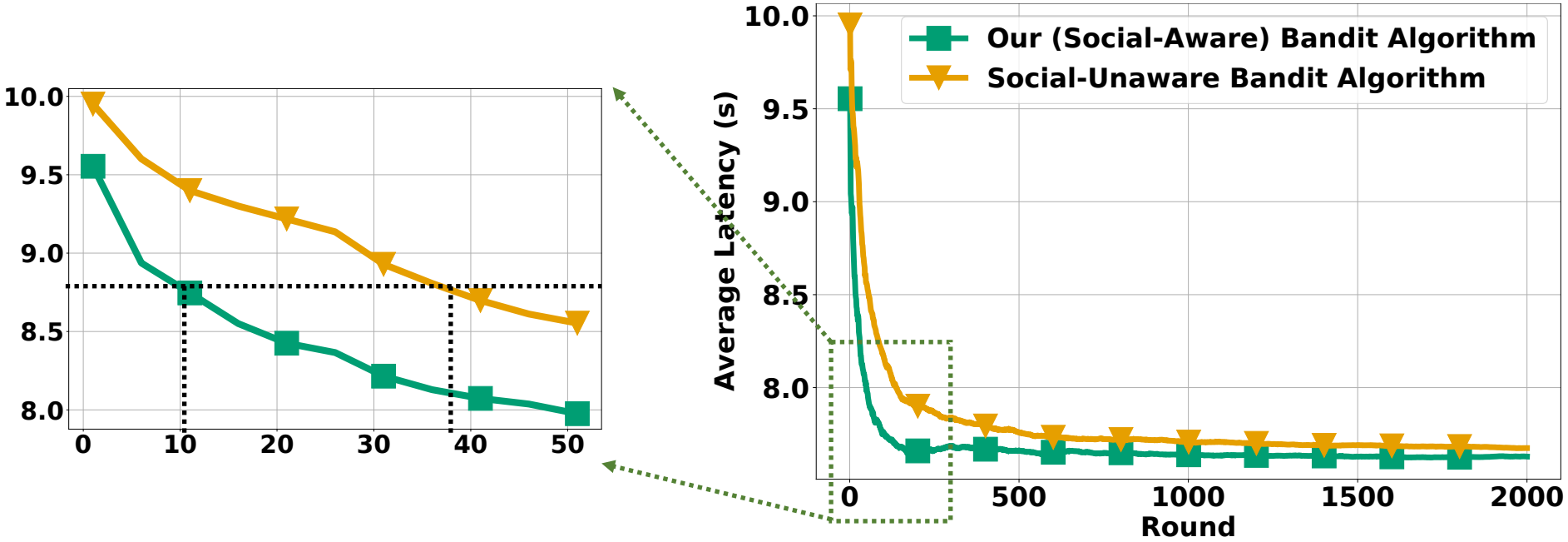
Sublinear guarantee: $V = \Omega(\sqrt{T/\log T}) \Rightarrow \mathrm{Reg}(T) = O(\sqrt{\log T/T})$

# Regret and Independence Number

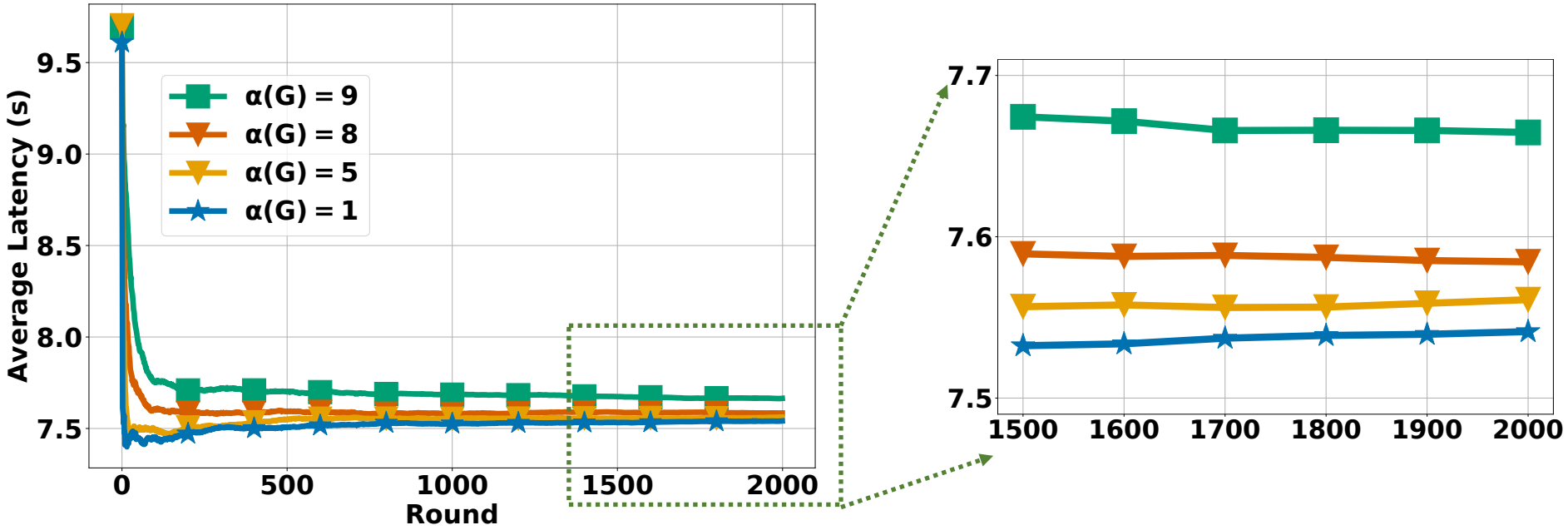


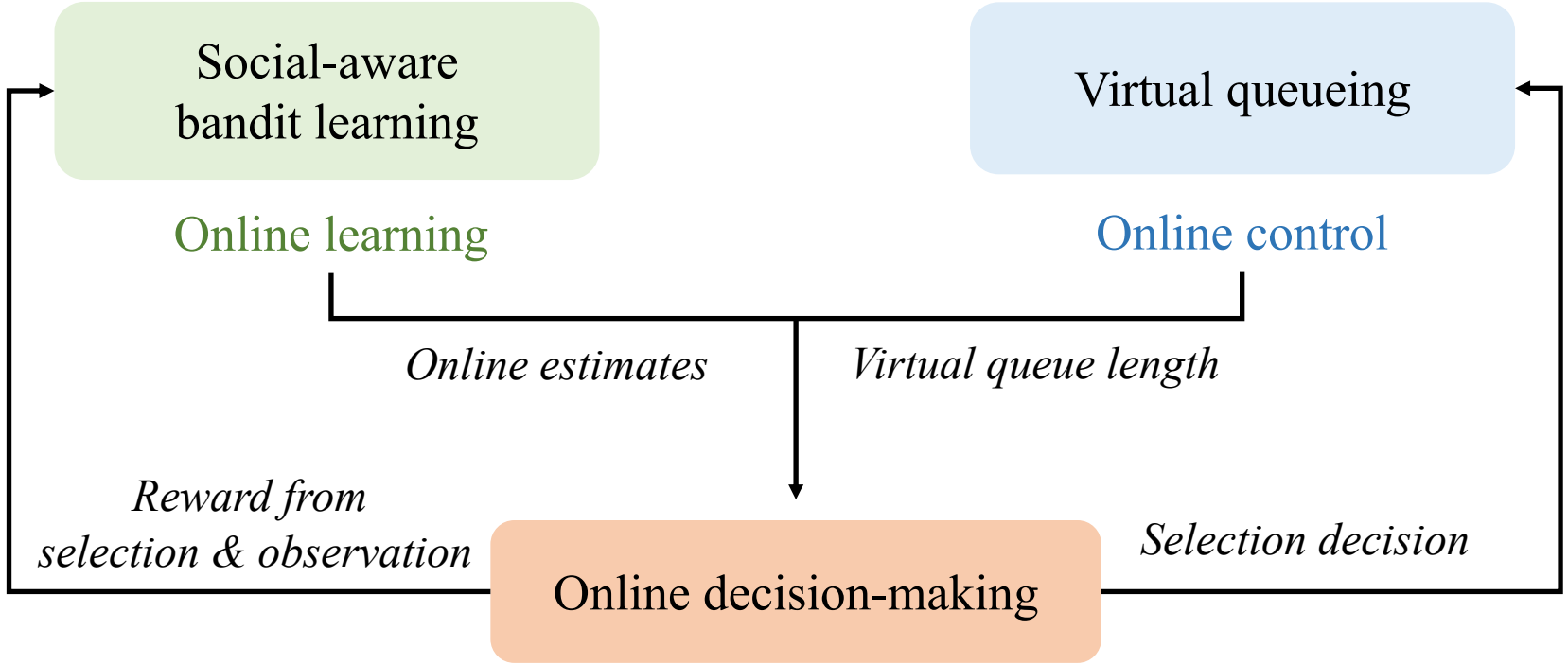
# Simulation Results: Latency



# Simulation Results: Independence Number



# Conclusion


Social-aware
bandit learning
Virtual queueing
Online learning
Online control
Online estimates
Virtual queue length
Reward from
selection & observation
Online decision-making
Selection decision

Thank
you